robot04 Autonomous Mapping: Experiments of 20 October 2026

Hector SLAM + Nav2 + frontier explorer on the Raspberry Pi 5

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Experiment log, 20 October 2026

Abstract

On 20 October 2026, robot04 (Hiwonder MentorPi T1, Raspberry Pi 5, MS200 lidar) ran **18 autonomous exploration runs** between 09:32 and 12:57, driving **153 m** in **32.5 min** of explorer time. Hector SLAM, Nav2 and the frontier explorer ran on the robot; the PC mapper app watched, and **8 maps** were saved from it. No code changed during the day: the runs differ in where they ran and how they ended. In a large open area (morning) 82 % of Nav2 goals were aborted and no run got back to its start. Four runs at 10:53 ended within 4 s on the explorer's low-battery rule. In a small walled enclosure (afternoon) 7 of 9 runs explored until no frontier was left and drove home; the last six runs had **no failed goal** (39 sent). The new mapping timer in the app was used on the last two runs and agrees with the explorer's own clock to within 12 s.

1 Setup

- **Robot:** robot04, Raspberry Pi 5, tracked base, Oradar MS200 360° lidar (10 Hz).
- **On the robot** (`systemctl -user start robot-explore`): Hector SLAM (0.05 m cells), Nav2 (NavFn, Regulated Pure Pursuit at 0.10 m/s), the frontier explorer (robot radius 0.15 m, minimum frontier 0.30 m, size weight 0.5, replan every 3 s, no-progress timeout 30 s, battery limit 7.0 V). The parameters were the same in every run (they are recorded in each log's `begin` event).
- **On the PC:** the robot04 mapper app shows the map, path, plan and frontiers; it saves the robot's `/map` as map_server `pgm`/`yaml` plus a JSON note.
- **Data** (all in `experiments/2026-10-20/data`): 18 explorer logs (`robot_logs/*.jsonl`, one event per goal plus a sample every few seconds), 17 maps the explorer saved on the robot (`robot_maps`) and the 8 maps saved from the PC (`pc_maps`).

A run starts when **Explore** is pressed and ends when the explorer reports *done* (no frontier left and back at the start, or a stop condition) or the operator stops it. The duration below is the time from *start* to that event; the distance is the explorer's sum of SLAM pose steps. Mapped area is the number of known cells (free or occupied) times 0.05^2 m^2; it includes everything the lidar saw, also beyond walls the robot could not pass.

2 Runs

The 18 runs fall into five sessions (table 1, fig. 1).

Table 1: Sessions. Failed goals: Nav2 aborted the goal or it timed out.

	Time	Conditions	Runs	Home	Failed / sent	Fail	Driven (m)	Completed runs (mean)
A	09:32–09:41	Large open area	4	0	65 / 79	82%	26.5	–
B	10:09–10:17	Long run, large area	1	0	2 / 16	12%	58.6	–
C	10:53–10:55	Low battery	4	0	0 / 5	0%	0.0	–
D	11:49–12:21	Bounded test area	5	4	10 / 34	29%	29.3	1.7 min, 6.9 m
E	12:38–12:57	Bounded test area, repeat	4	3	0 / 31	0%	38.7	2.3 min, 10.4 m

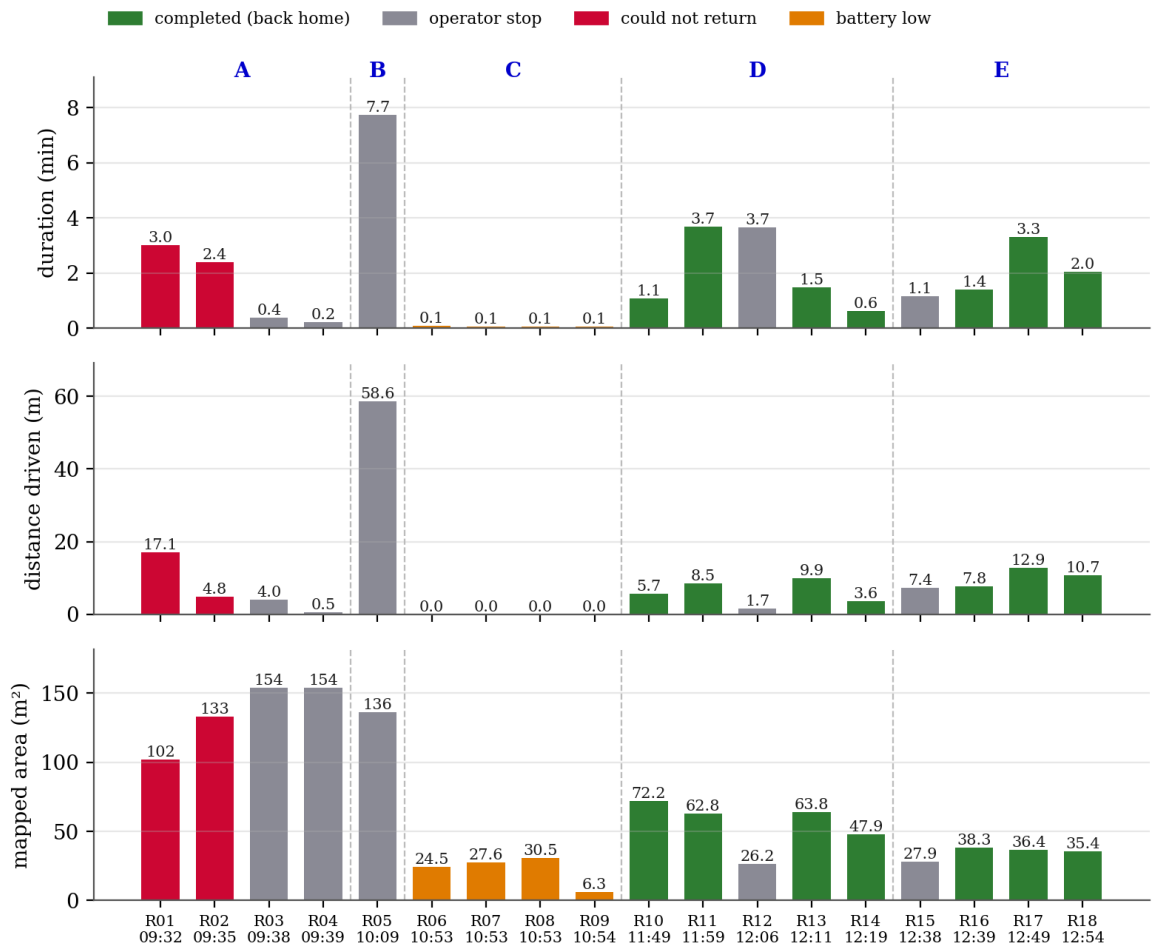


Figure 1: Every run: duration, distance driven and mapped area, coloured by how it ended. Sessions A–E are separated by dashed lines. In sessions A and C the SLAM map was not reset between runs, so their mapped area is cumulative.

Table 2: All runs. Reached: frontier goals Nav2 reached; seen: frontiers that disappeared while driving to them (the lidar saw them first); failed: Nav2 aborted or timed out.

Run	S.	Start	Time (s)	Dist. (m)	Area (m^2)	Sent	Reached	Seen	Failed	End
R01	A	09:32	181	17.1	102	41	1	12	29	return failed
R02	A	09:35	144	4.8	133	33	0	1	33	return failed
R03	A	09:38	23	4.0	154	1	0	0	0	operator stop
R04	A	09:39	12	0.5	154	4	0	0	3	log ends
R05	B	10:09	464	58.6	136	16	9	4	2	operator stop
R06	C	10:53	4	0.0	24	1	0	0	0	battery low → home
R07	C	10:53	4	0.0	28	1	0	0	0	battery low → home
R08	C	10:53	4	0.0	30	1	0	0	0	battery low → home
R09	C	10:54	4	0.0	6	2	1	0	0	battery low → home
R10	D	11:49	65	5.7	72	7	0	6	1	home
R11	D	11:59	221	8.5	63	12	3	5	4	home
R12	D	12:06	220	1.7	26	7	1	0	5	operator stop
R13	D	12:11	89	9.9	64	5	4	1	0	home
R14	D	12:19	37	3.6	48	3	1	2	0	home
R15	E	12:38	69	7.4	28	7	3	3	0	operator stop
R16	E	12:39	84	7.8	38	8	5	3	0	home
R17	E	12:49	199	12.9	36	8	8	0	0	home
R18	E	12:54	123	10.7	35	8	7	1	0	home

A, 09:32–09:41: large open area, goals aborted

The first two runs mapped 100–130 m^2 of a long area (fig. 5a), but Nav2 aborted 62 of 74 goals (status 6) and neither run could drive back to its start. Failed goals cluster at the far edge of the map and beyond walls seen through gaps; the explorer blacklisted each and sent the next, 41 goals in 3 minutes in R01. R03 was stopped by the operator; R04's log ends after 12 s. Frontier search took 80–150 ms per goal on this map (20–70 ms on the small afternoon maps). *Not yet known:* why Nav2 aborted so many goals here. Candidates to check are narrow passages under the 0.45 m wall clearance of the global costmap, and goals in cells that are free in Hector's map but inside the inflated costmap.

B, 10:09: one long run

R05 is the longest run of the day: 7.7 min, 58.6 m driven, 136.5 m^2 mapped, 9 goals reached and 2 failed (fig. 5b). It drove along a corridor about 15 m long and into the branch at its far end; the operator stopped it before it finished. Map `lab_2026-10-20_1008` was saved from this run.

C, 10:53–10:55: battery low

Four runs ended within 4 s of *start*: the explorer cancelled its first goal with "battery low" and returned home. The voltage samples in these logs are 7.02–7.06 V, just above the 7.0 V limit, but the check uses the instantaneous voltage, which dips under load when the motors start (fig. 3). R05 had ended at 7.07 V. By 11:49 the battery read 8.18 V again.

D, 11:49–12:21: small enclosure

From here on the robot ran inside a small walled enclosure, about 2.5 m across, whose layout changed between runs (fig. 6). The lidar sees out of the enclosure into the room, so 26–72 m^2 were mapped while the robot drove 1.7–9.9 m. Four of five runs finished (no frontier left, back home); 10 of 34 goals failed, outside the enclosure or against its walls, where the robot cannot drive. R12 was stopped after 3.7 min with 1.7 m driven and 5 failed goals.

E, 12:38–12:57: repeat runs, no failures

Three of four runs finished; R15 was stopped by the operator after 1.1 min. No goal failed in this session (31 sent), nor in R13 and R14 before it. R17 reached all 8 of its goals. The completed runs took 1.4–3.3 min, drove 7.8–12.9 m (mean speed 0.08 m/s, against a 0.10 m/s cruise speed) and needed 5–14 s to drive home after the last frontier.

3 Mapping time

Figure 2 shows how the mapped area grew in the completed runs. Each starts steeply (each goal opens a new view for a 360° lidar) and then levels off while the robot visits the remaining corners. In R11 the area stayed at 45–47 m^2 for more than 2 min before the last goals opened another 16 m^2.

The mapper app's new timer (it runs while SLAM poses arrive) was used for the last two maps (table 3): 211 s against the explorer's 199 s for R17, and 130 s against 123 s for R18. The timer reads longer because it also counts the seconds between starting SLAM and pressing *Explore*, and between the end of the run and saving the map. Use the explorer time for the time to map an area, and the app timer for the whole session.

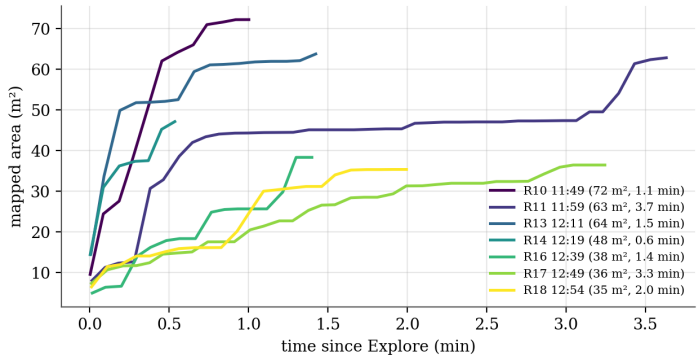


Figure 2: Mapped area against time since *Explore*, completed runs.

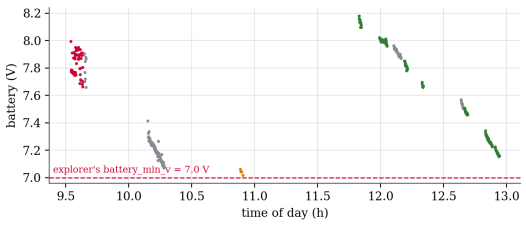


Figure 3: Battery voltage in every log sample, coloured by run outcome (orange: battery low).

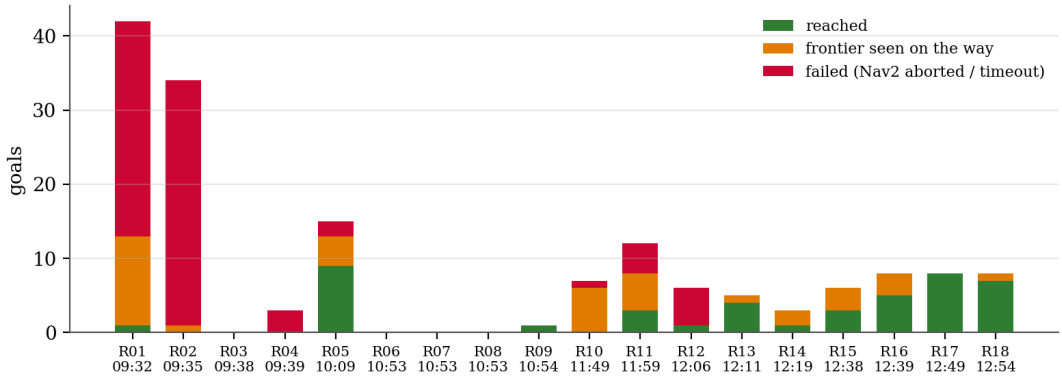


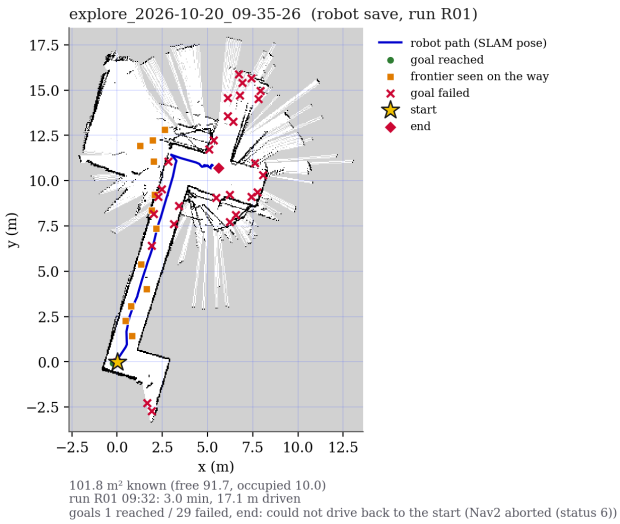
Figure 4: Goal outcomes per run.

4 Saved maps

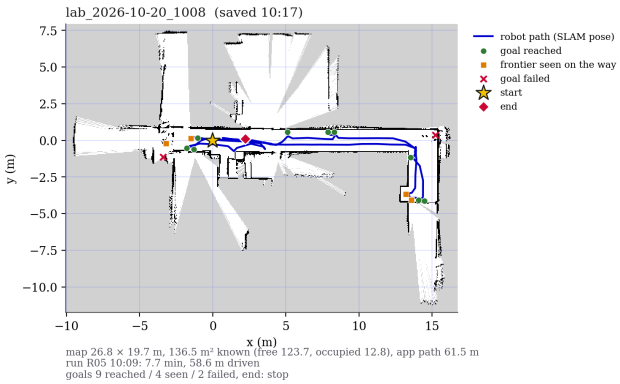
Table 3: Maps saved from the PC app (`lab_2026-10-20_*`). Known area = free + occupied. Timer: app timer / explorer time (s).

Map	Saved	Run	Extent (m)	Known (m^2)	Free	Occupied	Path (m)	Timer (s)
`1008`	10:17	R05	26.8×19.7	136.5	123.7	12.8	61.5	–
`1148`	12:03	R11	10.8×10.7	62.8	58.7	4.1	8.2	–
`1148_2`	12:10	R12	10.2×10.6	26.2	24.7	1.5	1.7	–
`1148_3`	12:13	R13	10.7×12.3	63.8	59.0	4.8	9.9	–
`1148_4`	12:20	R14	9.8×10.9	47.8	44.4	3.4	3.5	–
`1237_success`	12:41	R16	10.6×11.2	38.3	36.0	2.4	7.7	–
`1249_success2`	12:53	R17	10.6×10.3	36.4	33.9	2.6	12.8	211 / 199
`1249_success3`	12:56	R18	10.6×10.6	35.3	32.7	2.6	10.3	130 / 123

Every map is plotted with metric axes (x, y in the SLAM map frame, origin at the robot's start), the robot's path from the explorer's log and the goals. The figures are in `figures/maps/`.

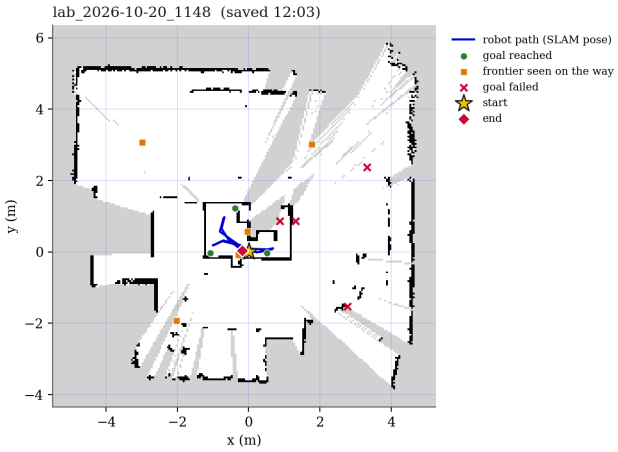


(a) R01 (session A), robot's own save.



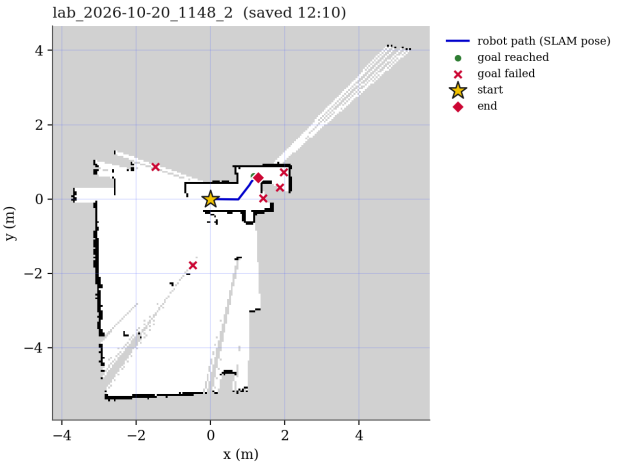
(b) R05 (session B).

Figure 5: Large-area runs.



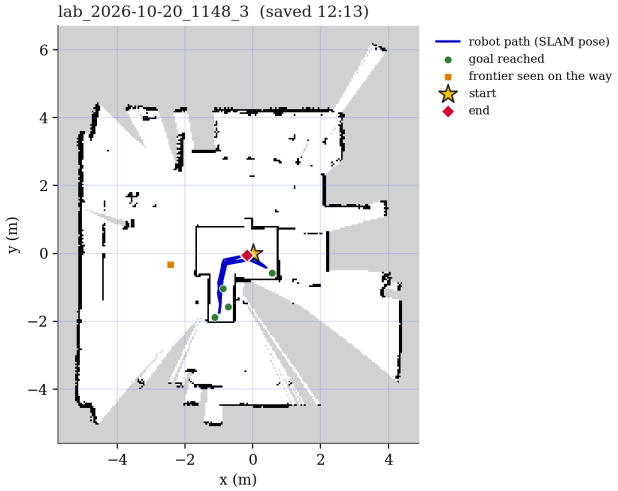
map 10.8 × 10.7 m, 62.8 m² known (free 58.7, occupied 4.1), app path 8.2 m
run R11 11:59: 3.7 min, 8.5 m driven
goals 3 reached / 5 seen / 4 failed, end: back at the start

(a) R11



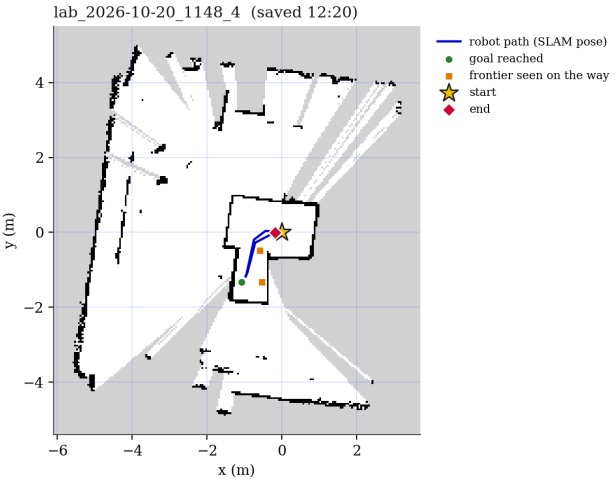
map 10.2 × 10.6 m, 26.2 m² known (free 24.7, occupied 1.5), app path 1.7 m
run R12 12:06: 3.7 min, 1.7 m driven
goals 1 reached / 0 seen / 5 failed, end: stop

(b) R12



map 10.7 × 12.3 m, 63.8 m² known (free 59.0, occupied 4.8), app path 9.9 m
run R13 12:11: 1.5 min, 9.9 m driven
goals 4 reached / 1 seen / 0 failed, end: back at the start

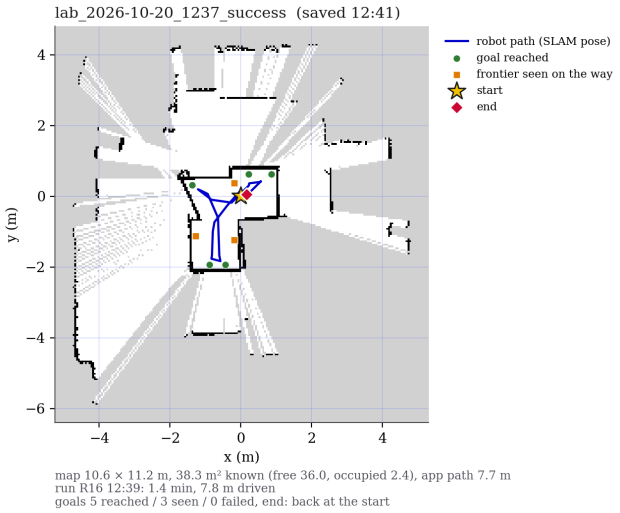
(c) R13



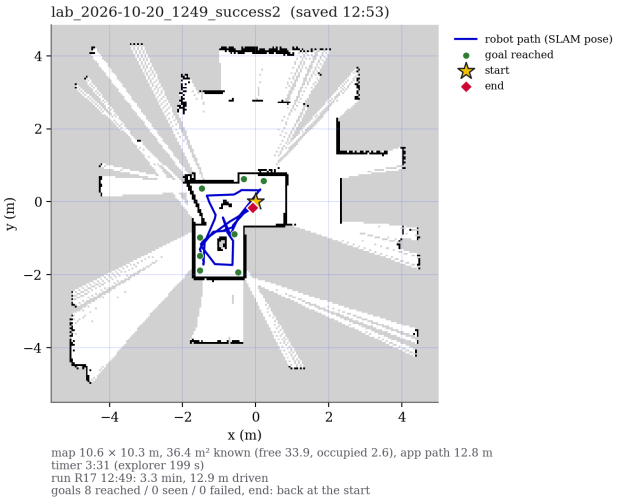
map 9.8 × 10.9 m, 47.8 m² known (free 44.4, occupied 3.4), app path 3.5 m
run R14 12:19: 0.6 min, 3.6 m driven
goals 1 reached / 2 seen / 0 failed, end: back at the start

(d) R14

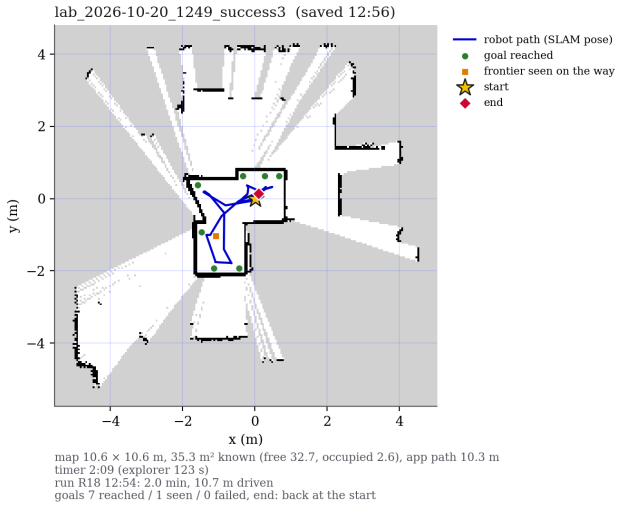
Figure 6: Session D maps.



(a) R16



(b) R17



(c) R18

Figure 7: Session E maps.

5 Findings and next steps

1. **The full loop works in a bounded area.** In the enclosure, 7 of 9 runs explored to the last frontier and drove home without help; the last six runs had no failed goal.
2. **Large open areas are not solved yet.** 82 % of goals aborted in session A and no run returned home. Next: record Nav2's abort reason and the costmap at each failed goal, and test with a smaller inflation radius.
3. **Frontiers the robot cannot reach cost time.** Goals outside the enclosure (session D) failed one by one. Next: drop frontiers that have no path in the global costmap before sending them.
4. **The battery rule trips on voltage dips.** Next: filter the voltage (for example a 5 s mean) before comparing it with 7.0 V, and log the minimum voltage.
5. **Timing.** Use the explorer's elapsed time for the time to map an area; the app timer now stores the session time with each map.

Reproduce. `python3 scripts/analyze_day.py && python3 scripts/make_tables.py`, then `pdflatex` this file twice in `report/`.